

Toliss_ParkBrake_Proper.lua

A FlyWithLua Script for Correct Parking Brake Operation
Thrustmaster TCA Quadrant Airbus Edition with ToLiss Aircraft in X-Plane 12

Author: Chris McKenna **Contact:** kb1sln@gmail.com

1. Overview

If you own a Thrustmaster TCA Quadrant Airbus Edition and fly ToLiss Airbus aircraft in X-Plane 12, you have likely discovered that the parking brake lever does not behave as expected. Moving the lever to the SET position either has no effect on the cockpit, or the brake sets for a split second and immediately releases on its own. This is not a defect in your hardware or your aircraft — it is a fundamental mismatch between how the TCA lever works at a hardware level and how X-Plane and ToLiss expect the parking brake to be controlled.

This script resolves that mismatch. It is a short FlyWithLua script that correctly reads the physical state of the parking brake lever and maps it to the ToLiss parking brake system every frame, giving you stable, realistic, lever-accurate brake control.

2. Understanding the TCA Parking Brake Lever

2.1 The Hardware Reality

The Thrustmaster TCA Quadrant Airbus Edition parking brake lever looks and feels like a two-position switch — you push it to SET and it stays there, you pull it back and it stays in PARK BRK. Physically it latches. But the way it communicates with your computer tells a different story.

The TCA parking brake lever is wired as a single virtual button. When the lever is in the SET position, that button reports as held down. When the lever is in the PARK BRK position, the button is not held. There is no second button for the released position — just one button that is either pressed or not pressed, depending on which position the lever is in.

This is well documented in the community. As noted in multiple X-Plane.org forum posts and the tomvand/xp-thrustmaster-tca GitHub repository, the TCA parking brake lever appears as button 20 on the device (button 19 in zero-based indexing), and crucially, the "parking brake off" button is absent from the default .joy configuration file — because there is no separate off button. The lever off state is simply the absence of the on button being pressed.

2.2 The Virtual Buttons Requirement

The TCA parking brake lever only registers as a button in X-Plane if virtual buttons are enabled in the Thrustmaster driver on Windows. Without this setting, the lever's axis position is not translated into button presses and X-Plane never sees any input from the lever at all.

Virtual buttons must be enabled in Windows Game Controllers before any software solution can work. See Section 5 for instructions.

2.3 The Button Indexing Challenge

X-Plane 12's joystick settings display button numbers starting at 1, making the parking brake lever appear as button 14 on a typical TCA setup. However, FlyWithLua's `button()` function uses a different numbering scheme — a global zero-based index that incorporates a per-device offset based on how many input devices are connected to your system and in what order Windows enumerates them.

This means the button number shown in X-Plane's joystick UI is not the same number you use in a Lua script. The correct global index must be discovered using a diagnostic script at runtime. For the author's system with a typical TCA + sidestick setup, the correct index is 19 — but this will vary by system. Section 5 provides a diagnostic script to find your number.

3. Why X-Plane's Built-In Bindings Cannot Handle This

X-Plane's native joystick binding system is designed around commands — discrete actions triggered when a button is pressed or released. When you bind a button to "Parking brake on" and another to "Parking brake off," X-Plane fires those commands at the moment the button transitions, not based on whether the button is currently held.

For a normal two-button setup this works fine. But the TCA has only one button for the entire lever. X-Plane has no way to distinguish "lever is physically in the SET position" from "lever was just moved to SET." It only sees a button transition event.

The ToLiss parking brake dataref (AirbusFBW/ParkBrake) expects a sustained value — 1 for on, 0 for off — not a momentary toggle. X-Plane's binding system cannot provide this sustained value from a single-button source without custom scripting. This is why a FlyWithLua script is necessary, and why this specific approach was developed.

4. How the Script Works

The script uses FlyWithLua's `do_every_frame()` function to run a check on every single rendered frame. On each frame, it reads the current physical state of the parking brake lever using FlyWithLua's `button()` function:

- If the lever is in the SET position, `button(19)` returns true and the script writes 1 to AirbusFBW/ParkBrake — brake ON.
- If the lever is in the PARK BRK position, `button(19)` returns false and the script writes 0 to AirbusFBW/ParkBrake — brake OFF.

Because the TCA lever physically latches — it stays in whatever position you put it — the button state is stable and sustained. The script reads that stable state every frame and keeps the ToLiss dataref in perfect sync with the physical lever position. The cockpit lever moves, the annunciators respond correctly, and the brake holds exactly as long as the physical lever is in the SET position.

The script also checks the aircraft ICAO code before writing to the dataref. It only activates for supported ToLiss Airbus types, so it will not interfere with other aircraft that use different parking brake systems.

The core logic is straightforward:

```
function TCA_ParkBrake_Update()
    if not IsTolissAirbus() then return end
    if button(PARK_BRAKE_BUTTON) then
        TL_ParkBrake = 1    -- Lever in SET position: brake ON
    else
        TL_ParkBrake = 0    -- Lever in PARK BRK position: brake OFF
    end
end

do_every_frame("TCA_ParkBrake_Update()")
```

5. Requirements

- **Simulator:** X-Plane 12
- **FlyWithLua NG Plugin:** The FlyWithLua NG (Next Generation) plugin must be installed. Download: <https://forums.x-plane.org/index.php?/files/file/82888-flywithlua-ng-next-generation-edition-for-x-plane-12/>
- **Hardware:** Thrustmaster TCA Quadrant Airbus Edition (with or without the Add-On modules)
- **ToLiss Aircraft:** Supported ICAO codes: A319, A320, A20N (A320neo), A321, A21N (A321neo), A339 (A330-900), A346 (A340-600)
- **Virtual Buttons:** Virtual buttons must be enabled in Windows — see Section 6

6. Installation

6.1 Enable Virtual Buttons on the TCA Quadrant

This step must be completed before anything else. Without it the parking brake lever will not register as a button in X-Plane.

1. Press the Windows Start button and type: joy
2. Click "Set up USB game controllers"
3. Double-click "TCA Q-Eng 1&2" in the device list
4. Click the "Properties" button
5. Check the box next to "Enable virtual buttons"
6. Click Apply, then OK

6.2 Install the Script

7. Copy Toliss_ParkBrake_Proper.lua into:
 <X-Plane 12 folder>/Resources/plugins/FlyWithLua/Scripts/
8. Start X-Plane 12 and load a supported ToLiss aircraft
9. Go to Plugins > FlyWithLua > Reload all Lua script files

6.3 Clear Any Existing X-Plane Button Binding

If you have previously assigned the parking brake lever to any command in X-Plane's joystick settings, that binding must be cleared. An active X-Plane binding and the Lua script will fight each other, resulting in erratic behaviour.

10. Go to Settings > Joystick in X-Plane 12
11. Select your TCA device and find the parking brake button in the button list
12. Click Edit next to it and set it to "Do nothing"
13. Click Done

7. Finding Your Button Number

The button number hardcoded in the script (19) is correct for the author's system. Because FlyWithLua's global button index depends on how many input devices are connected and how Windows orders them, this number may be different on your system. If the script does not work after installation, use the following diagnostic to find your correct button number.

7.1 Diagnostic Script

Create a new file called ParkBrake_Debug.lua with the following content:

```
-- ParkBrake_Debug.lua
function ShowButtonNumber()
```

```

    for i = 0, 1599 do
        if button(i) then
            logMsg("BTN: " .. i)
        end
    end
end

do_every_frame("ShowButtonNumber()")

```

7.2 Steps to Find Your Number

14. Save ParkBrake_Debug.lua to: <X-Plane 12>/Resources/plugins/FlyWithLua/Scripts/
15. In X-Plane 12, go to Plugins > FlyWithLua > Reload all Lua script files
16. Load any supported ToLiss aircraft
17. Push the parking brake lever to the SET position and hold it for a few seconds
18. Open <X-Plane 12>/Log.txt in a text editor
19. Search for lines beginning with "BTN:" — the number on those lines is your button index
20. Update the PARK_BRAKE_BUTTON value in Toliss_ParkBrake_Proper.lua with that number
21. Delete ParkBrake_Debug.lua from the Scripts folder
22. Reload FlyWithLua scripts and test

NOTE: The diagnostic script writes to the log every frame while any button is held. Remove it as soon as you have your number — leaving it running will cause log.txt to grow very rapidly and may slow X-Plane down.

8. Testing

With the script installed and the X-Plane binding cleared:

- Load a supported ToLiss aircraft at a gate or parking position
- Ensure the aircraft has power so the parking brake system is active
- Push the TCA parking brake lever to the SET position
- The cockpit parking brake lever should visibly move to the SET position
- The PARK BRK annunciator on the overhead panel should illuminate
- Pull the lever back to the PARK BRK (released) position
- The cockpit lever should return and the annunciator should extinguish

9. Troubleshooting

Nothing happens when I move the lever

- Verify that virtual buttons are enabled in Windows (Section 6.1)
- Run the diagnostic script to confirm your button number matches PARK_BRAKE_BUTTON in the script (Section 7)
- Confirm FlyWithLua was reloaded after installing the script

The brake sets but immediately releases

- An active X-Plane joystick binding on the parking brake button is conflicting with the script — clear it (Section 6.3)
- Check for any other parking brake Lua scripts in your FlyWithLua Scripts folder and remove them

The script works on one ToLiss aircraft but not another

- The aircraft ICAO code may not be in the supported list in the script
- Open the X-Plane developer console (Special menu) and note the PLANE_ICAO value for that aircraft
- Add the ICAO code to the IsTolissAirbus() function in the script

X-Plane warns about Runloop backed up in Log.txt

- This is caused by the diagnostic script running alongside the main script
- Remove ParkBrake_Debug.lua from the Scripts folder and reload FlyWithLua

10. Supported Aircraft

ICAO Code	Aircraft
A319	ToLiss Airbus A319
A320	ToLiss Airbus A320
A20N	ToLiss Airbus A320neo
A321	ToLiss Airbus A321
A21N	ToLiss Airbus A321neo
A339	ToLiss Airbus A330-900
A346	ToLiss Airbus A340-600

11. Full Script Listing

The complete script is reproduced below for reference. The script file itself is the authoritative version — install that file directly rather than copying from this document to avoid any formatting issues.

```
--
=====
=
-- Toliss_ParkBrake_Proper.lua
--
=====
=
-- Author:      Chris McKenna
-- Contact:     kb1sln@gmail.com
-- Version:     1.0
-- Date:        March 2026
-- Simulator:   X-Plane 12
-- Aircraft:    ToLiss A319, A320, A20N, A21N, A321, A339, A346
-- Hardware:    Thrustmaster TCA Quadrant Airbus Edition (with Add-On)
```

```

--
=====
=
--
-- DESCRIPTION:
-- This script correctly maps the Thrustmaster TCA Quadrant parking brake
-- lever to the ToLiss Airbus parking brake dataref (AirbusFBW/ParkBrake).
--
-- Without this script, the parking brake lever behaves erratically -- the
-- brake sets momentarily but immediately releases, resulting in a log
full
-- of rapid ON/OFF toggles and no visible lever movement in the cockpit.
--
-- WHY THE PROBLEM EXISTS:
-- The TCA parking brake lever generates only a single momentary virtual
-- button press when moved to the SET position. It does NOT generate a
-- separate button event when released back to the PARK BRK position.
-- X-Plane's built-in create_command mechanism uses on_begin (button down)
-- and on_end (button up) callbacks -- but because the lever's contact
-- is momentary, on_end fires almost instantly after on_begin, cancelling
-- the brake before it latches.
--
-- Additionally, X-Plane 12's new joystick UI does not have the old
-- "Buttons: Adv" tab. The button numbers shown in the UI are 1-indexed,
-- but FlyWithLua's button() function uses a global 0-based index that
-- includes a per-device offset. The correct index must be discovered
-- using a diagnostic script (see FINDING YOUR BUTTON NUMBER below).
--
-- HOW THIS SCRIPT WORKS:
-- Instead of create_command, this script uses do_every_frame() to poll
-- the physical state of the parking brake button on every rendered frame.
-- FlyWithLua's button(i) function returns true if the button is currently
-- held, false if not. Because the lever physically latches in the SET
-- position, button(19) remains true for as long as the lever is held
there.
-- This sustained true/false state is written directly to
AirbusFBW/ParkBrake
-- each frame, giving correct and stable behaviour.
--
-- REQUIREMENTS:
-- 1. X-Plane 12
-- 2. FlyWithLua NG (Next Generation) plugin
-- Download: https://forums.x-plane.org/index.php?/files/file/82888-flywithlua-ng-next-generation-edition-for-x-plane-12/
-- 3. Thrustmaster TCA Quadrant Airbus Edition
-- Virtual Buttons MUST be enabled in Windows:
-- Start Menu > "Set up USB game controllers" > TCA Q-Eng 1&2 >
-- Properties > check "Enable virtual buttons" > Apply
-- 4. ToLiss A319, A320, A321, A339, or A346
--
-- INSTALLATION:
-- 1. Copy this file to:
-- <X-Plane 12 folder>/Resources/plugins/FlyWithLua/Scripts/
-- 2. In X-Plane 12 Joystick settings, find button 14 (Park Brk) and
-- set it to "Do nothing" -- this prevents the old binding from

```

```

--      fighting the script.
-- 3. In X-Plane 12, go to Plugins > FlyWithLua > Reload all Lua script
files
-- 4. Test by pushing the parking brake lever to SET -- the cockpit lever
--      should move and the PARK BRK annunciator should illuminate.
--
-- FINDING YOUR BUTTON NUMBER:
-- The button number 19 below was found using a diagnostic script and is
-- correct for the author's system. It may differ on other systems
depending
-- on how many input devices are connected and in what order. If the
script
-- does not work, use the following diagnostic to find your button number:
--
--      function ShowButtonNumber()
--          for i = 0, 1599 do
--              if button(i) then
--                  logMsg("BTN: " .. i)
--              end
--          end
--      end
--      do_every_frame("ShowButtonNumber()")
--
-- Save this as a separate .lua file in the Scripts folder, reload
FlyWithLua,
-- push the parking brake lever to SET, then check X-Plane 12/Log.txt for
-- lines starting with "BTN:". The number reported is your button index.
-- Update PARK_BRAKE_BUTTON below with that value, then remove the
-- diagnostic script.
--
-- SUPPORTED AIRCRAFT ICAO CODES:
-- A319 - ToLiss Airbus A319
-- A320 - ToLiss Airbus A320
-- A20N - ToLiss Airbus A320neo
-- A21N - ToLiss Airbus A321neo
-- A321 - ToLiss Airbus A321
-- A339 - ToLiss Airbus A330-900
-- A346 - ToLiss Airbus A340-600
--
-- LICENSE:
-- Free to use, modify, and distribute. Credit appreciated but not
required.
-- Shared with the X-Plane community. No warranty expressed or implied.
--
=====
=

dataref("TL_ParkBrake", "AirbusFBW/ParkBrake", "writable")
dataref("PLANE_ICAO", "sim/aircraft/view/acf_ICAO", "readonly")

-- The global FlyWithLua button index for the TCA parking brake lever.
-- This was found using a diagnostic script (see header comments above).
-- If this does not work on your system, follow the FINDING YOUR BUTTON
NUMBER
-- instructions above to determine the correct value for your setup.

```



```

local PARK_BRAKE_BUTTON = 19

-- Returns true if the currently loaded aircraft is a supported ToLiss
Airbus.
-- The script only writes to AirbusFBW/ParkBrake when a ToLiss aircraft is
-- loaded, avoiding conflicts with other aircraft that use different
datarefs.
function IsTolissAirbus()
    if PLANE_ICAO == nil then return false end
    local icao = string.upper(tostring(PLANE_ICAO))
    return (icao == "A319" or icao == "A320" or icao == "A20N" or
            icao == "A21N" or icao == "A321" or icao == "A339" or icao ==
"A346")
end

-- Called every rendered frame by do_every_frame().
-- Reads the current physical state of the parking brake lever button and
-- writes the corresponding value (1=ON, 0=OFF) to the ToLiss parking brake
-- dataref. Because the TCA lever physically latches in the SET position,
-- button(PARK_BRAKE_BUTTON) remains true for the entire time the lever is
-- held there, giving stable and correct brake state.
function TCA_ParkBrake_Update()
    if not IsTolissAirbus() then return end
    if button(PARK_BRAKE_BUTTON) then
        TL_ParkBrake = 1
    else
        TL_ParkBrake = 0
    end
end

do_every_frame("TCA_ParkBrake_Update()")

```

12. License and Credits

This script is free to use, modify, and distribute. Credit is appreciated but not required. It is shared freely with the X-Plane community with no warranty expressed or implied. If you improve it or adapt it for additional aircraft, please consider sharing your changes back to the community.

Author: Chris McKenna **Contact:** kb1sln@gmail.com